

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration: 1 Hour
Total Marks:50

Annexure A

Scenario Based Assignment

Code of Conduct:

I Beffy A Shanika(192511387) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

beffy

Annexure A

Scenario Based Assignment

1. Scenario : A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is highly dynamic—roads may suddenly become inaccessible, and weather conditions affect travel cost. The drone has access to: A heuristic estimate (straight-line distance), Partial real-time updates about blocked paths and Variable movement costs depending on terrain and weather. However, due to urgency, the drone must quickly decide routes with minimum delay and risk.

Tasks:

- i. Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty
- ii. Explain how A* would handle the same situation, including how it balances cost and heuristic
- iii. Analyze the impact of heuristic accuracy on both algorithms
- iv. Discuss how dynamic changes (blocked paths) affect search performance
- v. Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety

2. Scenario: A smart city implements an AI system to optimize traffic signal timings across hundreds of intersections. The objective is to minimize Total waiting time, Fuel consumption and Traffic congestion. The system must continuously adjust signals based on traffic flow. However the search space is extremely large, traffic patterns change dynamically and the system may get stuck in locally optimal solutions

Tasks. Formulate this as an optimization problem and explain why traditional search is inefficient

- i. Explain how Hill Climbing would work in this scenario and identify its limitations
- ii. Discuss issues like local maxima, plateaus, and ridges with real-world interpretation
- iii. Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations
- iv. Recommend the best approach and justify your choice based on scalability and adaptability

3. Scenario: An autonomous rover is deployed on Mars to explore unknown terrain. It must: navigate safely to collect samples, avoid hazards (craters, rocks), operate with limited sensing capability and make decisions without a complete map. The rover updates its knowledge as it explores, making this a real-time decision-making problem.

Tasks:

- i. Explain why this problem requires an online search agent instead of offline search
- ii. Analyze the challenges of partial observability and dynamic environments
- iii. Describe how the rover builds and updates its internal model
- iv. Compare online search with uninformed and informed search strategies
- v. Justify the most suitable approach considering uncertainty, adaptability, and safety

4. **Scenario:** A university is designing an AI system to automatically generate exam timetables. The system must satisfy multiple constraints: No student should have overlapping exams, Limited number of exam halls, Certain subjects must be scheduled before others, Faculty availability constraints, Special accommodations for some students, The complexity increases as the number of courses and students grows.

Tasks :

Formulate the problem as a Constraint Satisfaction Problem (CSP)

- i. Clearly define variables, domains, and constraints with explanation
- ii. Explain how backtracking search works in this scenario
- iii. Discuss how constraint propagation (e.g., forward checking) improves efficiency
- iv. Analyze real-world challenges and justify the best strategy for scalability

5. **Scenario:** Strategic Game AI for Real-Time Decision Making (Minimax & Alpha-Beta Pruning) A gaming company is developing an AI opponent for a real-time strategy game. The AI must compete against human players, Make decisions within strict time limits, Evaluate complex game states with multiple possible moves, The decision tree is extremely large, making computation expensive.

Tasks:

- i. Explain how the Minimax algorithm models this problem
- ii. Analyze the computational challenges of large game trees
- iii. Explain how Alpha-Beta pruning improves efficiency with detailed reasoning
- iv. Design an evaluation function and justify the factors considered
- v. Recommend a strategy for balancing optimality and real-time performance

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

1. AI-Powered Drone (Greedy Best-First Search & A*)

i. Greedy Best-First Search

- Chooses the path with the lowest heuristic (estimated distance).
- Very fast for emergency situations.
- May choose unsafe or costly routes because it ignores actual travel cost.

ii. A* Search

- Uses $f(n) = g(n) + h(n)$.
- $g(n)$: Actual travel cost.
- $h(n)$: Estimated distance to goal.
- Finds safer and more optimal routes by considering both cost and heuristic.

iii. Impact of Heuristic Accuracy

- Accurate heuristic → Faster and better results.
- Poor heuristic → More search time and less efficient paths.
- A* is less affected than Greedy because it also considers actual cost.

iv. Effect of Dynamic Changes

- Blocked paths require replanning.
- Greedy may repeatedly choose blocked routes.
- A* adapts better by recalculating the lowest-cost path.

v. Recommendation

A* is the best choice because it provides optimal, safe, and reliable paths while adapting to changing conditions.

2. Smart City Traffic Optimization (Hill Climbing & GA/SA)

i. Optimization Problem

Objective:

- Minimize waiting time.
- Minimize fuel consumption.
- Reduce traffic congestion.

Traditional search is inefficient because the search space is very large.

ii. Hill Climbing

- Starts with one solution.
- Continuously improves the signal timings.
- Stops when no better neighboring solution exists.

Limitation: Gets stuck in local optimum.

iii. Local Maxima, Plateaus, Ridges

- Local Maxima: Best nearby solution but not global best.
- Plateau: Many equal-quality solutions.
- Ridge: Good solution exists but difficult to reach.

iv. Simulated Annealing / Genetic Algorithm

- Simulated Annealing: Accepts worse solutions occasionally to escape local maxima.
- Genetic Algorithm: Combines multiple solutions using selection, crossover, and mutation.

v. Recommendation

Genetic Algorithm is recommended because it is scalable, adaptable, and produces high-quality solutions.

3. Mars Rover (Online Search Agent)

i. Why Online Search?

The rover does not have a complete map and must make decisions while exploring.

ii. Challenges

- Partial observability.
- Unknown obstacles.
- Dynamic environment.
- Limited sensors.

iii. Internal Model

The rover:

- Stores explored locations.
- Updates obstacle information.
- Learns safer routes over time.

iv. Online vs Offline Search

Online Search

Explores while moving

Adapts to changes

Suitable for unknown environments

Offline Search

Complete map needed

Fixed plan

Suitable for known environments

v. Recommendation

Online Search Agent is the best choice because it is adaptive, safe, and works well in uncertain environments.

4. University Exam Timetable (Constraint Satisfaction Problem)

i. CSP Formulation

- Variables: Exams.
- Domains: Available dates and time slots.
- Constraints:
 - No overlapping exams.
 - Limited halls.
 - Faculty availability.
 - Subject order.
 - Special accommodations.

ii. Backtracking Search

- Assigns exams one by one.
- If a conflict occurs, returns to the previous assignment and tries another option.

iii. Forward Checking

- Removes invalid choices early.
- Detects conflicts before they happen.
- Reduces search time.

iv. Best Strategy

Backtracking with Forward Checking is the best approach because it efficiently handles large scheduling problems.

5. Game AI (Minimax & Alpha-Beta Pruning)

i. Minimax

- Assumes both players play optimally.
- Maximizes AI's score and minimizes the opponent's score.

ii. Challenges

- Huge search tree.
- High computation time.
- Slow decision-making.

iii. Alpha-Beta Pruning

- Removes branches that cannot affect the final decision.
- Reduces the number of nodes evaluated.
- Produces the same optimal result faster.

iv. Evaluation Function

Factors:

- Player score.
- Resources.
- Health.
- Territory control.
- Remaining moves.

v. Recommendation

Use Minimax with Alpha-Beta Pruning because it provides near-optimal decisions with much faster execution, making it suitable for real-time strategy games.